Physical Chemistry (I)

Lecture 06

First Law of Thermodynamics

In the Last Class

- Real gas
  - van der Waals equation
  - van der Waals equation and virial expansion
  - Beyond van der Waals
- The first law
  - The first law of thermodynamics
  - Work and heat

Last class

van der Waals Equation

- Repulsion: decrease of the total volume

$$V \to V - nb$$

$$p = \frac{nRT}{V - nb}$$

- Attraction: decrease of the pressure

$p \sim$ (frequency of collisions) × (force of each particle)

decreases as $n/V \uparrow$ decreases as $n/V \uparrow$

$$p = \frac{nRT}{V - nb} - a\frac{n^2}{V^2}$$

van der Waals and Virial

- Virial expansion

$$pV_m = RT\left(1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \cdots\right)$$

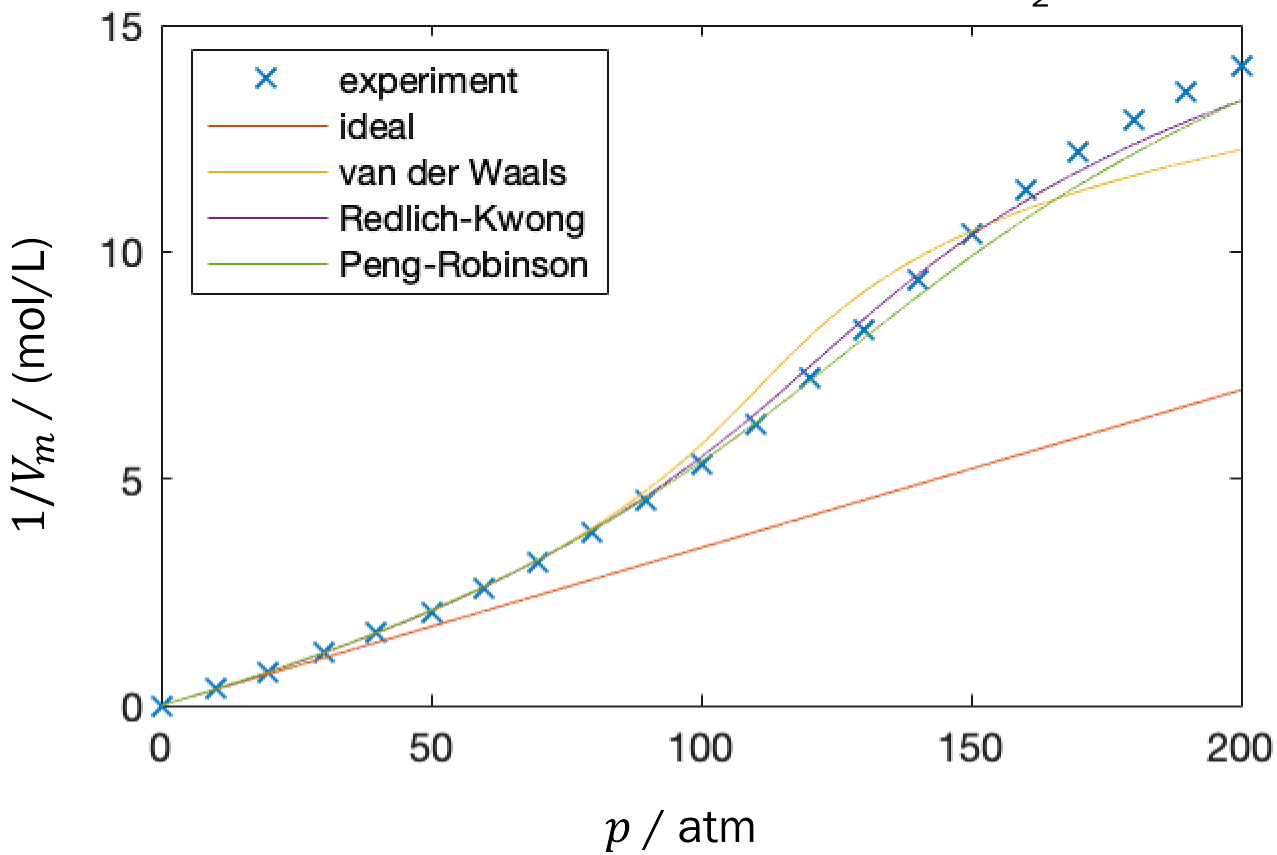
- van der Waals equation

$$p = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

$$pV_m = RT\left[1 + \left(b - \frac{a}{RT}\right)\frac{1}{V_m} + \frac{b^2}{V_m^2} + \cdots\right]$$

Beyond van der Waals

Data: CO_2 at 350 K



Last class

First Law of Thermodynamics

"The internal energy of an isolated system is constant."

- Heat and work are equivalent ways of changing the internal energy of a system

$$\Delta U = q + w$$

$$\underbrace{\mathrm{d}U}_{\text{small change of } U} = \underbrace{\delta q}_{\text{small } q} + \underbrace{\delta w}_{\text{small } w}$$

Last class

Calculation of Expansion Work

$$\delta w = -p_{\mathrm{ex}}\mathrm{d}V$$

- Total work for a change from state i to state f

$$w = \int_{\text{state } i}^{\text{state } f} \delta w = \int_{\text{state } i}^{\text{state } f} (-p_{\mathrm{ex}}\mathrm{d}V) = -\int_{V_i}^{V_f} p_{\mathrm{ex}}\,\mathrm{d}V$$

1. Expansion against constant external pressure
2. Reversible expansion

Calculation of Heat

$$\delta q = C\mathrm{d}T$$

- Total heat for a change from state i to state f

$$q = \int_{\text{state } i}^{\text{state } f} \delta q = \int_{\text{state } i}^{\text{state } f} (C\mathrm{d}T) = \int_{T_i}^{T_f} C\,\mathrm{d}T$$

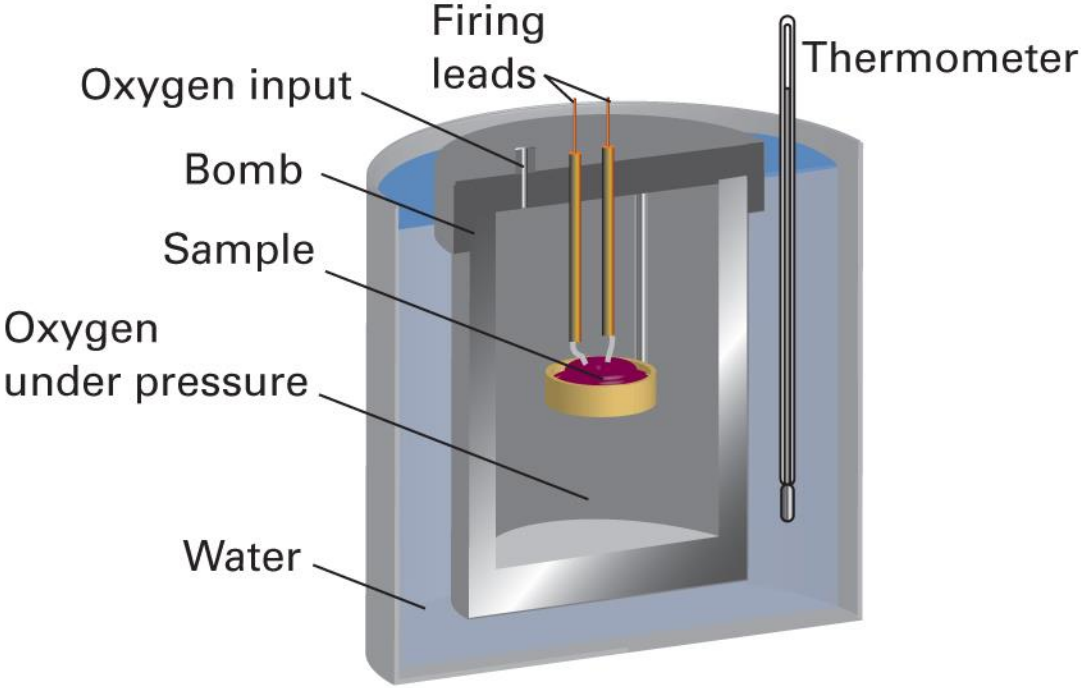
Heat Capacity

- Molar heat capacity $C_m = C/n$
- Specific heat capacity $C_s = C/m$

- Heat capacity depends on the constraint
  - C_V: heat capacity at constant volume ($C_{V,m}$, $C_{V,s}$)
  - C_p: heat capacity at constant pressure ($C_{p,m}$, $C_{p,s}$)

Topic 2A

Calorimetry: Heat Measurement



(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 11th ed., Figure 2A.8)

First Law Revisited

$$\mathrm{d}U = \delta q + \delta w$$

$$\mathrm{d}U = \delta q + \delta w_{\mathrm{exp}} + \delta w_{\mathrm{add}}$$

$$\mathrm{d}U = C\mathrm{d}T - p_{\mathrm{ex}}\mathrm{d}V + \delta w_{\mathrm{add}}$$

For $\delta w_{\mathrm{add}} = 0$,

$$\mathrm{d}U = C\mathrm{d}T - p_{\mathrm{ex}}\mathrm{d}V$$

At Constant V

$$\mathrm{d}U = C\mathrm{d}T - p_{\mathrm{ex}}\mathrm{d}V$$

At constant volume, $\mathrm{d}V = 0$ and $C = C_V$

$$\mathrm{d}U = C_V\mathrm{d}T$$

The change in internal energy is

$$\Delta U = \int_{\text{state } i}^{\text{state } f} \mathrm{d}U = \int_{\text{state } i}^{\text{state } f} (C_V\mathrm{d}T) = \int_{T_i}^{T_f} C_V\,\mathrm{d}T$$

At Constant V

$$\Delta U = \int_{T_i}^{T_f} C_V \, \mathrm{d}T$$

If C_V is constant,

$$\Delta U = \int_{T_i}^{T_f} C_V \, \mathrm{d}T = C_V \int_{T_i}^{T_f} \mathrm{d}T = C_V\left(T_f - T_i\right) = C_V \Delta T$$

Since $\Delta U = q + w = q$ at constant volume,

$$q = C_V \Delta T$$

Not in Atkins

(Monatomic) Perfect Gas

$$U = \frac{3}{2} nRT$$

$$\mathrm{d}U = \frac{3}{2} nR\mathrm{d}T$$

$$\mathrm{d}U = C\mathrm{d}T - p_{\mathrm{ex}}\mathrm{d}V$$

$$\frac{3}{2} nR\mathrm{d}T = C\mathrm{d}T - p_{\mathrm{ex}}\mathrm{d}V$$

Not in Atkins

Perfect Gas at Constant V

$$\frac{3}{2}nR\mathrm{d}T = C\mathrm{d}T - p_{\mathrm{ex}}\mathrm{d}V$$

At constant volume, $\mathrm{d}V = 0$ and $C = C_V$

$$\frac{3}{2}nR\mathrm{d}T = C_V\mathrm{d}T$$

Hence,

$$C_V = \frac{3}{2}nR$$

Now, get ready for some math…

Topic 2D

State and Path Functions

- State function: a property that depends only on the current state of a system and is independent of its history

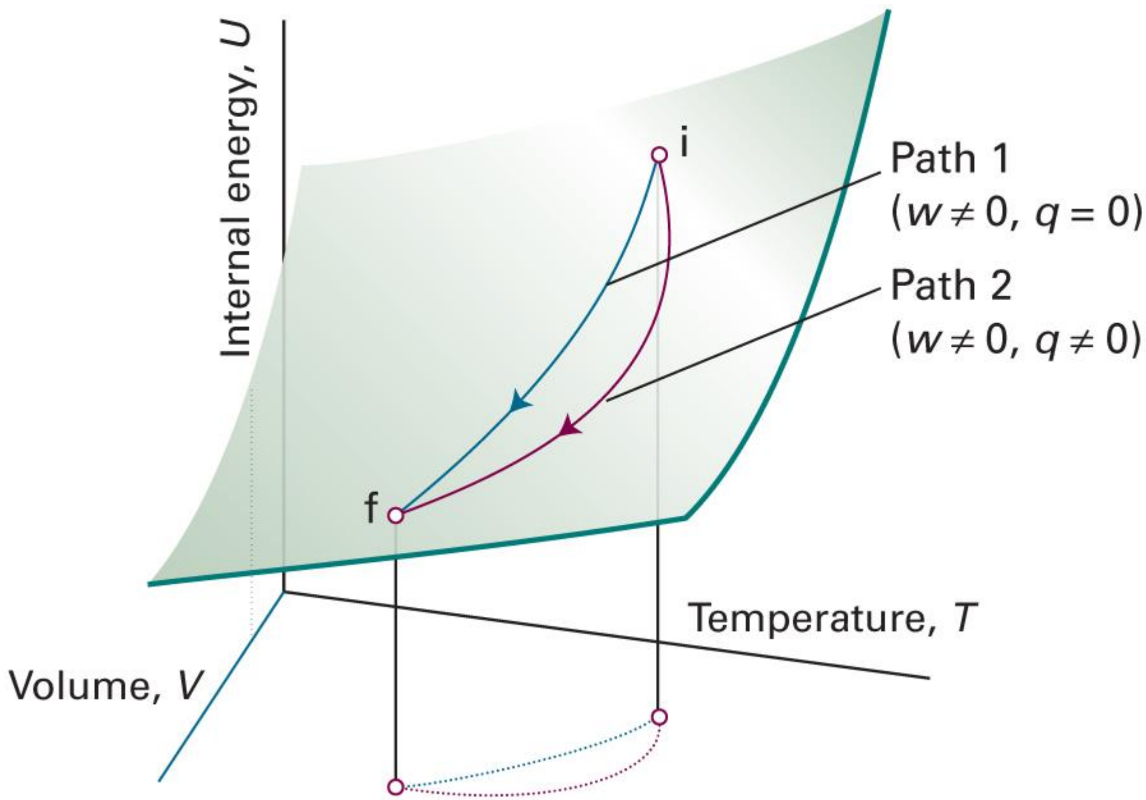
  e.g. internal energy, pressure, volume, temperature

- Path function: a property with a value that depends on the path between two states

  e.g. heat, work

Topic 2D

State and Path Functions



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 2D.1)

Related Mathematics

- Exact differential: an infinitesimal quantity that, when integrated, gives a result that is independent of the path between the initial and final states.

$$\Delta U = \int_{\text{state } i}^{\text{state } f} \mathrm{d}U$$

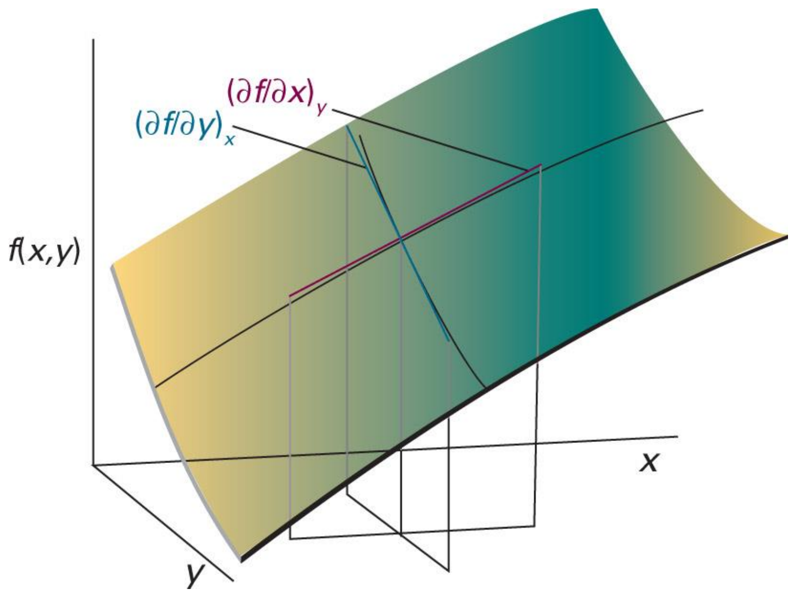
- Inexact differential: an infinitesimal quantity that, when integrated, gives a result that depends on the path between the initial and final states.

$$q = \int_{\text{state } i,\, \text{path}}^{\text{state } f} \delta q$$

Review

Partial Derivatives

"the slope of the function with respect to one of the variables, all the other variables being held constant"



$$\left(\frac{\partial f}{\partial x}\right)_y = \lim_{\Delta x \to 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$$

$$\left(\frac{\partial f}{\partial y}\right)_x = \lim_{\Delta y \to 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}$$

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., the chemist's toolkit 9, Sketch 1)

Differential

If f is a function of x and y, the differential df is

$$\mathrm{d}f = \left(\frac{\partial f}{\partial x}\right)_y \mathrm{d}x + \left(\frac{\partial f}{\partial y}\right)_x \mathrm{d}y$$

when df is an exact differential.

State Variables

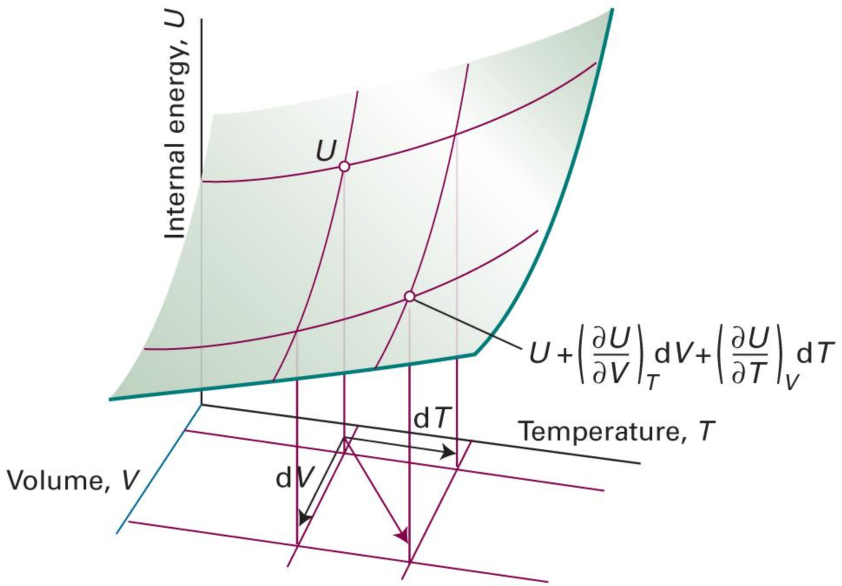
- A state is defined by p, V, n, T
- Typically n is fixed
- Equation of state: p is constrained by V and T
- We can just use V and T to define a state

$$U = U(V, T)$$

Changes in Internal Energy

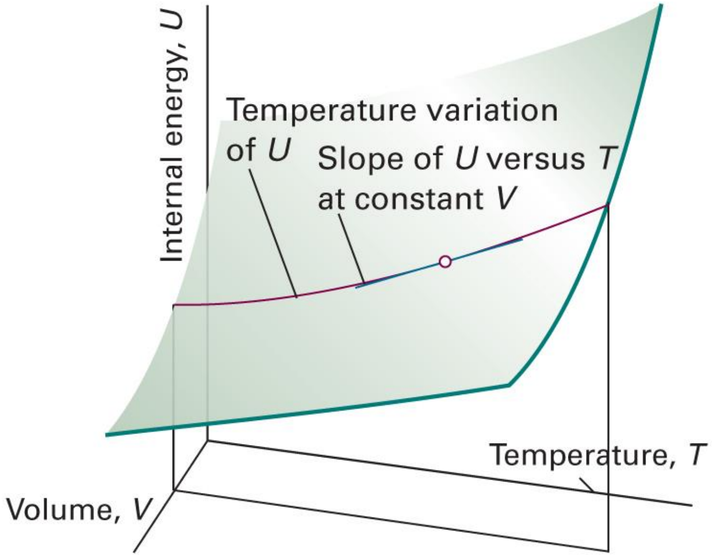
For $U = U(V, T)$,

$$\mathrm{d}U = \left(\frac{\partial U}{\partial V}\right)_T \mathrm{d}V + \left(\frac{\partial U}{\partial T}\right)_V \mathrm{d}T$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 2D.2)

Meaning of $(\partial U/\partial T)_V$

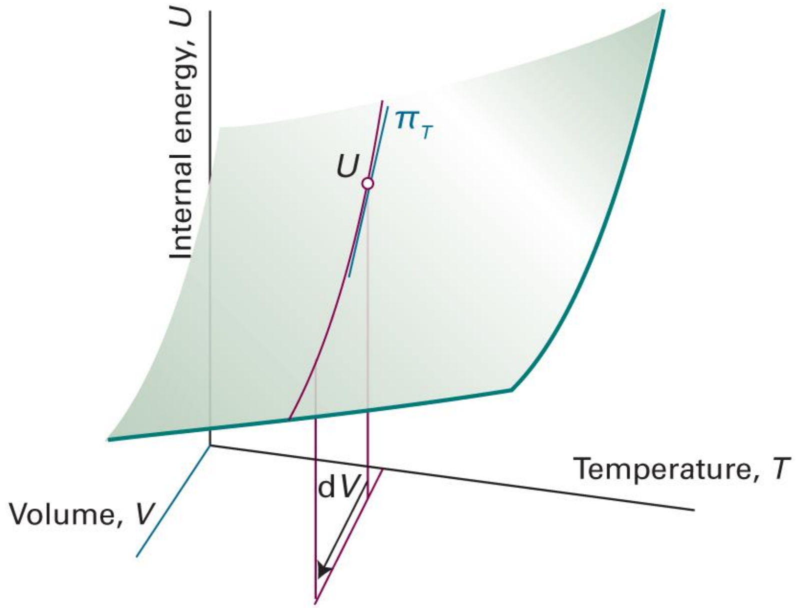


$$\mathrm{d}U = C_V \mathrm{d}T \text{ (at constant } V\text{)}$$

$$\left(\frac{\partial U}{\partial T}\right)_V = C_V$$

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 2A.10)

Meaning of $(\partial U/\partial V)_T$



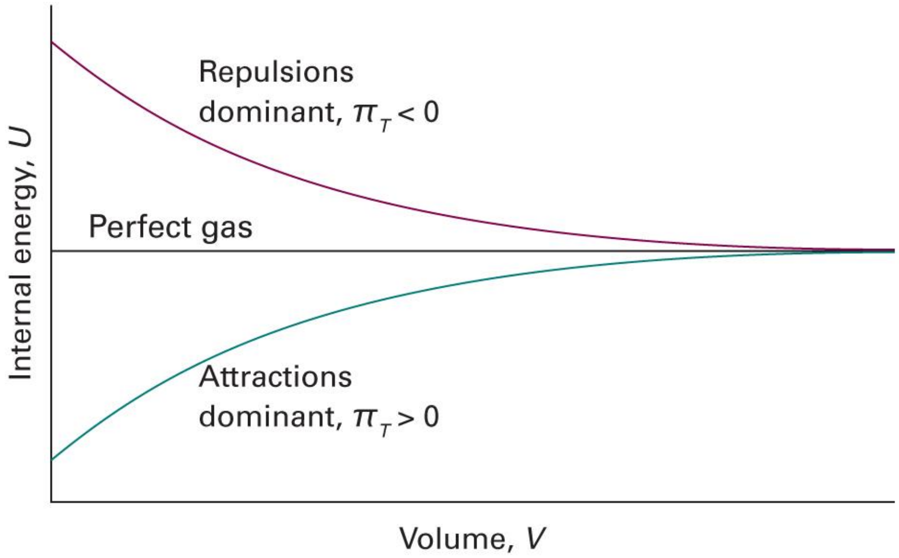
variation of the internal energy of a substance

as its volume is changed at constant temperature

(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 2D.3)

Internal Pressure

$$\pi_T = \left(\frac{\partial U}{\partial V}\right)_T$$



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 2D.4)

Use of Differentials

$$\mathrm{d}U = \underbrace{\left(\frac{\partial U}{\partial V}\right)_T}_{\pi_T} \mathrm{d}V + \underbrace{\left(\frac{\partial U}{\partial T}\right)_V}_{C_V} \mathrm{d}T$$

$$\mathrm{d}U = \pi_T \mathrm{d}V + C_V \mathrm{d}T$$

At constant pressure,

$$\left(\frac{\partial U}{\partial T}\right)_p = \pi_T \left(\frac{\partial V}{\partial T}\right)_p + C_V$$

Response Functions

- Expansion coefficient

$$\alpha = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_p$$

- Isothermal compressibility

$$\kappa_T = -\frac{1}{V}\left(\frac{\partial V}{\partial p}\right)_T$$

Some Data

	α / (10^{-4} K^{-1})	κ_T / (10^{-6} bar^{-1})
Liquids:		
Benzene	12.4	90.9
Water	2.1	49.0
Solids:		
Diamond	0.030	0.185
Lead	0.861	2.18

General Expression

$$\left(\frac{\partial U}{\partial T}\right)_p = \pi_T \left(\frac{\partial V}{\partial T}\right)_p + C_V$$

Using the definition of the expression coefficient α,

$$\left(\frac{\partial U}{\partial T}\right)_p = \alpha \pi_T V + C_V$$

For a perfect gas, $\pi_T = 0$ and

$$\left(\frac{\partial U}{\partial T}\right)_p = C_V = \left(\frac{\partial U}{\partial T}\right)_V$$

Example 2D.2

Derive an expression for the expansion coefficient of a perfect gas.

$$\alpha = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_p$$

For a perfect gas, $V = nRT/p$

$$\left(\frac{\partial V}{\partial T}\right)_p = \left(\frac{\partial}{\partial T}\left(\frac{nRT}{p}\right)\right)_p = \frac{nR}{p}$$

Thus,

$$\alpha = \frac{1}{V}\frac{nR}{p} = \frac{nR}{nRT} = \frac{1}{T}$$

Summary for Chapter 3 (1)

First Law of Thermodynamics

$$\mathrm{d}U = \delta q + \delta w$$

Work	$\delta w = -p_{\mathrm{ex}}\mathrm{d}V + \delta w_{\mathrm{add}}$ • Constant-p change • Reversible change
Heat	$\delta q = C\mathrm{d}T$ • Constant-V change $C_V = (\partial U/\partial T)_V$

Thermodynamics	Mathematics
State function	Exact differential
Path function	Inexact differential

Differential form of U

$$\mathrm{d}U = \left(\frac{\partial U}{\partial V}\right)_T \mathrm{d}V + \left(\frac{\partial U}{\partial T}\right)_V \mathrm{d}T$$

- Internal pressure π_T
- Response functions α, κ_T